

updated: 4/26

Curriculum Vitae: Chad McKell

ABOUT

Position	Research Engineer / Ph.D. Candidate
Company	Sigma Connectivity, on assignment at Meta
Address	10301 Willows Road NE, Redmond, WA 98052
Email	cmckell@meta.com
Website	chadmckell.com
Languages	English (native), Spanish (B2), French (A2/B1)
Code	Python, C++/CUDA, MATLAB, LaTeX
Research	My research spans mathematical modeling and numerical simulation of physical systems, with an emphasis on acoustics and general wave physics. Recent projects include conformal symmetry for exterior wave problems; differential forms for non-spherical harmonic decomposition; and variational principles for elastic wave modeling. Past projects include underwater acoustic sensing and optical standing-wave trapping.

EDUCATION

9/19–(exp. 9/26)	University of California San Diego , Ph.D. in Computer Music Research area: Computational Acoustics and Applied Mathematics Dissertation: <i>Conformal Symmetry for Exterior Wave Problems</i> Advisors: Albert Chern (CSE) and Miller Puckette (Music; IRCAM)
9/16–10/17	University of Edinburgh , M.S. in Acoustics and Music Technology
8/09–12/15	Wake Forest University , M.S. in Physics
6/02–8/09	Brigham Young University , B.S. in Biophysics

EMPLOYMENT

9/24–	Sigma Connectivity, on assignment at Meta , Research Engineer
3/24–6/24	University of California San Diego , Teaching Assistant in Computer Science
6/23–2/24	Meta, Reality Labs Research , Research Scientist Intern
8/21–3/22	Meta, Reality Labs Research , Research Intern
9/19–6/23	University of California San Diego , Teaching Assistant/Researcher in Music
7/18–7/19	Applied Research in Acoustics , R&D Scientist
5/18–5/18	Moog Music (self-employed) , Audio Software Developer
4/17–9/17	Lofelt, acquired by Meta in 2022 (self-employed) , Acoustics Researcher
10/14–8/16	J.P. Morgan/Neovest (via ConsultNet) , Software Development Engineer in Test
8/12–12/12	University of North Carolina School of the Arts , Adjunct Instructor of Physics
9/09–9/11	Wake Forest University , Teaching Assistant in Physics
9/08–6/09	Brigham Young University , Tutorial Lab Assistant in Physics
8/07–3/09	Brigham Young University , Research Assistant in Philosophy

RESEARCH ACTIVITIES

9/24–	Sigma Connectivity, on assignment at Meta , Research Engineer Redmond, Washington. Affiliation: Reality Labs Research Audio. Research topics: <i>spatial audio</i> . Research summary: Build novel spatial audio technology for Reality Labs Research at Meta. Supervisor: Sebastian Prepelitã. Team Lead: Ravish Mehra. I work onsite at Meta as a vendor employed by Sigma Connectivity.
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RESEARCH ACTIVITIES CONT.

- 9/19–9/24 **University of California San Diego**, Ph.D. Student
La Jolla, California. Affiliations: Department of Music, Center for Visual Computing. Research topics: *computational acoustics*, *applied mathematics*. Research summary: developed mathematical and computational methods for solving exterior wave problems, with applications in virtual acoustics. Committee: Albert Chern (co-chair, CSE), Miller Puckette (co-chair, Music; IRCAM), Melvin Leok (Mathematics), Shahrokh Yadegari (Music), Stefan Bilbao (Univ. of Edinburgh; IRCAM), Sebastian Prepelîță (Meta).
- 6/23–2/24 **Meta, Reality Labs Research**, Research Scientist Intern
Redmond, Washington. Affiliation: Reality Labs Research Audio. Research topics: *spatial audio*. Research summary: Built novel spatial audio technology for Reality Labs Research at Meta. Supervisor: Sebastian Prepelîță. Team Lead: Ravish Mehra.
- 8/21–3/22 **Meta, Reality Labs Research**, Research Intern
La Jolla, California. Research description: see above.
- 7/18–7/19 **Applied Research in Acoustics**, R&D Scientist
Culpeper, Virginia. Research topics: *underwater acoustic sensing*, *acoustic beamforming*. Research summary: developed physics-based signal processing algorithms for object detection and classification in naval sonar systems. Team Lead: Jonathan Botts.
- 1/17–8/17 **University of Edinburgh**, Master's Student
Edinburgh, Scotland. Affiliations: Reid School of Music, Edinburgh College of Art. Research topics: *computational acoustics*, *elastodynamics*, *numerical simulation*. Research summary: developed numerical simulations of structural acoustic vibrations for commercial haptic devices. My thesis was partially funded by Lofelt, a Berlin-based haptic feedback company acquired by Meta in 2022. Advisor: Stefan Bilbao.
- 1/10–9/13 **Wake Forest University**, Master's Student
Winston-Salem, North Carolina. Affiliation: Department of Physics. Research topics: *optical trapping*, *laser beam characterization*, *fluid dynamics*. Research summary: simulated, fabricated, and analyzed standing-wave optical traps for Brownian particle tracking. Advisor: Keith Bonin.
- 8/07–8/09 **Brigham Young University**, Undergraduate Student
Provo, Utah. Affiliation: Department of Physiology and Developmental Biology. Research topics: *biophysics*, *scanning probe microscopy*. Research summary: investigated structural effects of isoflurane on artificial lipid bilayers using atomic force microscopy. Advisor: David Busath.

TEACHING EXPERIENCE

As Instructor

UNCSA
SCI 1100 General Physics. Fall 2012 (1 term).

TEACHING EXPERIENCE CONT.

As TA

UCSD

CSE 291	Physics Simulation. Spring 2024 (1 term).
MUS 5	Sound in Time. Spring 2020 (1 term).
MUS 6	Electronic Music. Fall 2020 (1 term).
MUS 15	Popular Music: David Bowie. Winter 2021 (1 term).
MUS 15	Popular Music: Video Game Music. Winter 2020 (1 term).
MUS 171	Computer Music I. Winter 2022 (1 term).
MUS 172	Computer Music II. 2021–2022 (2 terms).

WFU

PHY 113	General Physics I (Mechanics). 2009–2011 (4 terms).
PHY 114	General Physics II (E&M). Fall 2010 (1 term).

As Tutor

BYU

PHSCS 105	General Physics 1 (Mechanics). 2008–2009 (2 terms).
PHSCS 106	General Physics 2 (E&M). Winter 2009 (1 term).
PHSCS 121	Principles of Physics 1 (Mechanics). 2008–2009 (2 terms).
PHSCS 123	Principles of Physics 2 (Waves/Thermo). W/Sp 2009 (2 terms).
PHSCS 220	Principles of Physics 3 (E&M). W/Sp 2009 (2 terms).

PUBLICATIONS

Manuscripts in Progress

- (1) **C. McKell**, M. Nabizadeh, S. Wang, and A. Chern, “Wave simulations in infinite spacetime”.

Simulating wave propagation on an infinite domain has been a long-standing computational challenge. Conventional approaches to this problem only produce wave simulations on a small subset of the infinite domain. Using the fact that wave propagation on an infinite Minkowski spacetime is equivalent to wave propagation on a bounded Minkowski spacetime under a Kelvin-like transformation, we simulate wave propagation on the entire infinite domain using a finite discretization of the bounded domain with no additional loss of accuracy from the transformation.

- (2) **C. McKell**, *Conformal Symmetry for Exterior Wave Problems*, Ph.D. Dissertation, University of California San Diego. Defense planned for Fall 2025. Advisors: Albert Chern and Miller Puckette.

This work presents novel geometric methods for handling open exterior boundaries in wave simulations. First, I discuss extensions to the reflectionless discrete perfectly matched layer for the wave and Helmholtz equations. Then, I demonstrate that the conformal invariance of the wave equation under a Kelvin transform in Minkowski spacetime allows one to convert an infinite domain problem into a bounded domain problem that can be solved using standard numerical methods with no additional loss of accuracy introduced by the transform. I conclude by discussing parallel computing strategies for the geometric boundary models, applications in architectural acoustics and binaural audio, and future work in obstacle boundary flattening.

PUBLICATIONS CONT.

Journal Articles

- (3) **C. McKell** and K. Bonin, “Optical corral using a standing-wave Bessel beam,” *Journal of the Optical Society of America B*, Vol. 35, No. 8, 1910–1920, 2018.

Conference Proceedings

- (4) **C. McKell**, “Sonification of optically-ordered Brownian motion,” In Proceedings of the International Computer Music Conference (ICMC), Utrecht, Netherlands, September 2016.

Master’s Theses

- (5) **C. McKell**, *Real-Time Physical Modeling for Haptic Feedback Rendering*, Master’s Thesis, University of Edinburgh, Acoustics and Audio Group, 2017. Advisor: Stefan Bilbao.
- (6) **C. McKell**, *Confinement and Tracking of Brownian Particles in a Bessel Beam Standing Wave*, Master’s Thesis, Wake Forest University, Department of Physics, 2015. Advisor: Keith Bonin.

Technical Reports

- (7) **C. McKell**, H. Conley, and D. Busath, “AFM study of structural changes in supported planar DPPC bilayers containing general anesthetic isoflurane,” Brigham Young University, Paper 827, 2010.